

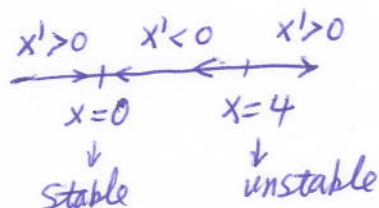
Instructions: There are totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Given the following initial value problem:

$$\frac{dx}{dt} = x^2 - 4x, \quad x(t_0) = x_0.$$

(a) Find all critical points of the given autonomous differential equation and then determine whether it is stable or unstable for each of them.

Solution: $x^2 - 4x = 0 \Rightarrow x(x-4) = 0$
 $x = 0$ or $x = 4 \rightarrow$ two critical points



(b) Solve the differential equation explicitly for $x(t)$ in term of t .

Solution: $\int \frac{1}{x^2 - 4x} dx = \int dt$
 $\int \left(-\frac{1}{4}\right) \left(\frac{1}{x} - \frac{1}{x-4}\right) dx = t + C$
 $\ln \left| \frac{x}{x-4} \right| = -4t + C$
 $\frac{x}{x-4} = Ae^{-4t}$
 $x(t_0) = x_0 \Rightarrow \frac{x_0}{x_0-4} = Ae^{-4t_0}$
 $\Rightarrow A = \frac{x_0 e^{4t_0}}{x_0 - 4}$

Thus
 $\frac{x}{x-4} = \frac{x_0}{x_0-4} e^{-4(t-t_0)}$
 $x(x_0-4) = x_0(x-4) e^{-4(t-t_0)}$
 $x(x_0-4 - x_0 e^{-4(t-t_0)}) = -4x_0 e^{-4(t-t_0)}$
 $x(t) = \frac{-4x_0 e^{-4(t-t_0)}}{x_0-4 - x_0 e^{-4(t-t_0)}}$